Tertiary Infotech Academy Pte Ltd

Lab 10 — Filter and Slice Finance Data Safely

Python Programming for Finance · TGS-2026065049 · LO5

Scenario

A portfolio manager asks for bank-sector positions above a materiality threshold and a date-windowed price series.

Goal

A filtered positions file and labelled versus positional slice comparison.

Tools

Pandas, CSV

Step-by-step

1. Load portfolio.csv and market_prices.csv.

2. Use loc with a boolean mask for Bank sector and Value above threshold.

3. Use loc for an inclusive date-label slice.

4. Use iloc for a positional sample and compare endpoints.

```
python starter.py
```

5. Sort without mutating the original DataFrame.

6. Save filtered_positions.csv and run tests.

```
python -m unittest test_solution.py
```

Acceptance criteria

The filter returns only qualifying bank positions; loc and iloc row counts match their documented boundary rules.

Evidence

Capture the executed output, validation result, generated file or chart, and a one-sentence finance interpretation.